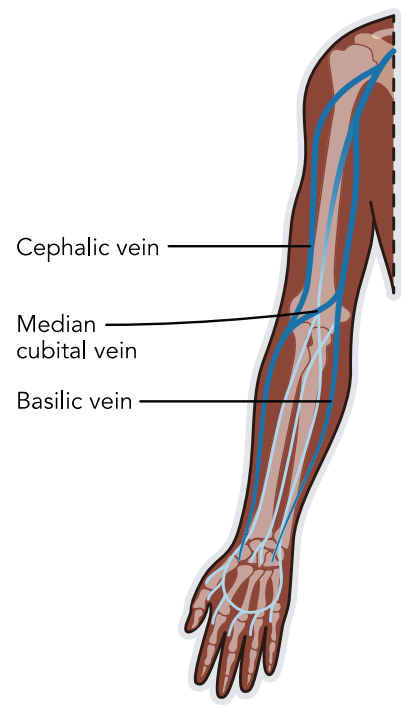
Besides the expiration date, tubes should be examined to confirm that there are not cracks or splits in the tube or stopper. It is also necessary to confirm that the tubes will hold enough blood for the tests to be performed. Each blood test requires a certain amount of blood to help assure the accuracy of the test. For example, if the tests require 11 mL of blood and the tube you need comes only in 6-mL sizes, you will use two tubes for that collection.

Site Selection in the Antecubital Region

Blood collected by venipuncture uses the veins in the ***antecubital*** region of the arm and the dorsal part of the hand. The antecubital region is located slightly below the bend of the elbow, anterior to the elbow joint.

Knowing and being able to find the best vein for venipuncture is one of the most important steps of performing this type of blood collection. Venipuncture can be performed in the antecubital region of the arm and the dorsal part of the hand. In rare cases and with permission from the provider, the foot or ankle may be used.

The first choice for a routine venipuncture is the antecubital region of the arm. To help place your patient in position, once they are seated or reclined, ask them to extend their arm out straight, palm up. This will reveal the antecubital region, located slightly below the bend of the elbow. This is the area on the other side of the elbow about midway from the hand to the shoulder. If you are unsure where this region is on your patient, place your thumb on their elbow and wrap your hand around the arm. Where your fingers land is the antecubital area on your patient.



antecubital. The inside of the elbow; location of the veins used in venipuncture.

median cubital. Vein near the middle of antecubital region; first choice for venipuncture.

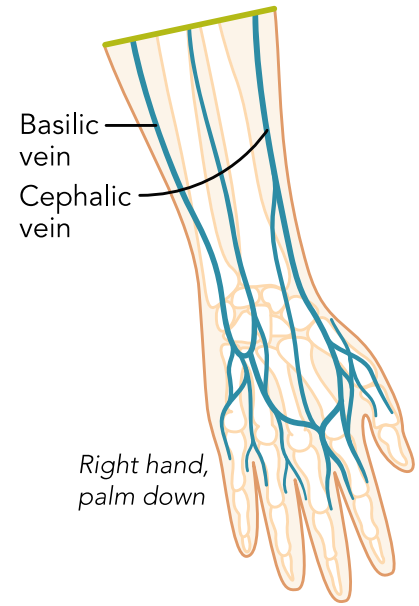
cephalic. Vein located in the antecubital, near the outside of the arm; second choice for venipuncture.

basilic. Vein located in the antecubital near the body; the last choice for venipuncture.

The veins typically used for venipuncture in this area include the ***median cubital***, ***cephalic***, and ***basilic***. The median cubital vein is the best choice for venipuncture as it is located near the antecubital region. Its location also helps it stay better anchored in place in the arm. The cephalic vein is located closer to the outer side of the arm and is the second-best choice. The basilic vein is located closer to the inner side of the body. Because this vein is located close to the median nerve and brachial artery and rolls more easily, a needle miss in this area could harm the patient. This vein should only be used if you cannot find a vein suitable for venipuncture in either arm or hand.

Site Selection in the Dorsal Part of the Hand

The veins in some patients may be very difficult to find in their antecubital region, or the ones that you can find may not be acceptable choices for venipuncture. When this occurs, the next best choice for venipuncture is in the dorsal part of the hand. This region is found by asking your patient to extend their hand, palm down. While there are designated veins that are accessed in antecubital region, the veins in the hand are chosen by their ability to safely withstand a venipuncture. Choose veins that are well anchored and large enough to allow a needle to enter and have the fewest other anatomical structures under the vessel. Avoid crooked veins or ones that feel too close to a tendon. The acceptable area for venipuncture is the entire surface of the hand.



Never perform a venipuncture on the inside of the wrist. There is a risk of injuring a nerve or tendon in this area, and it should be avoided for blood collection no matter how visible or assessable the veins in this location appear.

When there are no choices available and the need for the blood outweighs the risks, leg or ankle blood collections may be performed but only with the permission from the ordering personnel or the patient's provider. Confirm with your facility to determine who is qualified to perform this procedure. When this site is the only available location and you have permission, use veins that are straight and well anchored in the leg or ankle. Cleanse the area and bandage well due to the increased risk of infection. Do not perform leg or ankle venipunctures on patients who have been diagnosed with diabetes mellitus.

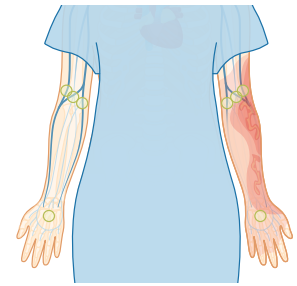
Age and physical condition must be taken into consideration when preparing to perform a venipuncture or other blood collection. Some of these factors are physical and affect the supplies you choose. Others influence patient communication and positioning, while some determine the amount of blood that can be safely collected.

The age of your patient can impact factors such as positioning and communication. Older adults may have physical issues such as arthritis, difficulty hearing, or visual impairments. While these conditions may not have a direct impact on the blood collection itself, they do impact your interactions with the patient. Communication and mobility issues can affect people of all ages and are not always immediately apparent. Some simple considerations can have a large impact on how successful blood collections are performed. These factors may not always change the supplies or procedures but may call for extra attention to your communication with the patient.

4.1 CHALLENGE

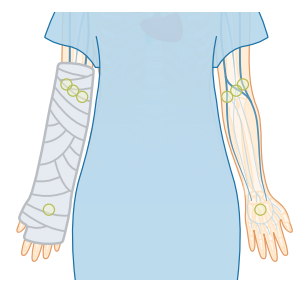
You are preparing to collect blood from three patients. For each patient, assess the potential blood collection sites. Keep in mind that each patient has their own complication that can affect where you should perform the collection.

1. Which of the following is the complication: burn covering most of the left arm and hand with some scarring?



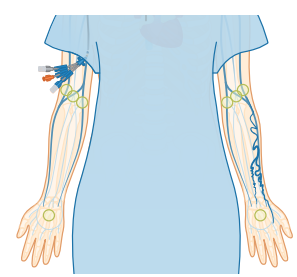
   A. 1: Right cephalic - not visible or palpable
   B. 2: Right median cubital - not visible or palpable
   C. 3: Right basilic - visible but not palpable
   D. 4: Right hand - visible and palpable
   E. 5: Left hand - visible and palpable
   F. 6: Left basilic - not visible or palpable
   G. 7: Left median cubital - visible and palpable
   H. 8: Left cephalic - visible and palpable

2. Complication: CVAD located in the upper right forearm with some tortuous veins present on the lower left forearm and hand



   A. 1: Right cephalic - visible and palpable
   B. 2: Right median cubital - visible but not palpable
   C. 3: Right basilic - visible and palpable
   D. 4: Right hand - visible and palpable
   E. 5: Left hand - visible and palpable
   F. 6: Left basilic - palpable but not visible
   G. 7: Left median cubital - visible and palpable
   H. 8: Left cephalic - visible and palpable

3. Complication: Cast on the right arm



   A. 1: Right cephalic – not visible or palpable
   B. 2: Right median cubital - not visible or palpable
   C. 3: Right basilic - not visible or palpable
   D. 4: Right hand - not visible or palpable
   E. 5: Left hand - visible and palpable
   F. 6: Left basilic - visible and palpable
   G. 7: Left median cubital - not visible or palpable
   H. 8: Left cephalic - visible and palpable

4.1 CHALLENGE KEY

1. D. Since the patient has significant burns covering most of their left arm and hand, this side should be avoided. The right cephalic and median cubital are not visible or palpable and the right basilic is only visible but not palpable. Therefore, the best option for this venipuncture would be the dorsal side of the right hand.
2. G. Since the patient has a CVAD on the right side, this side should be avoided. While the patient does have some tortuous veins in the left forearm, they are located distally and will not interfere with use of the antecubital space. Since both the median cubital and cephalic are visible and palpable on the left, the left median cubital should be used since this vein has the lowest chance of pain and complications.
3. H. The veins in the right arm and hand are not visible or palpable due to the cast. While the left cephalic and left basilic are both visible and palpable, the left cephalic should be used since it has the lower chance of pain and complications.